

Machine Translation and Implementation of Transformers

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Abstract

This report covers the topics presented in the “Introductions to Large Language Models” course taught by Tanmoy Chakraborty and Soumen Chakrabarti, available on NPTEL. As a graded assignment, the report follows a set of seven questions, and accompanies a code base available in the PLACEHOLDER directory. Each question will be quoted in full, followed by explanations and code listings where appropriate.

1 Sequence-to-Sequence Model

- 1.(a) Implement a Transformer-based sequence-to-sequence model in PyTorch for machine translation (English $\rightarrow$ French).
- 1.(b) Explain the encoder-decoder structure, attention mechanism, and training process.
- 1.(c) Provide code snippets and discuss evaluation metrics such as BLEU score.

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